

Vatsal Gandhi

+1 (646) 542-2485 | gandhivatsal76@gmail.com | [linkedin.com/in/vats-gandhi](https://www.linkedin.com/in/vats-gandhi) | github.com/Vats10987

EDUCATION

Master of Science in Computer Science

New York Institute of Technology, USA

Sep 2025 – May 2027

GPA: 3.95

Courses: Artificial Intelligence, Machine Learning, Big Data Analytics, Adv. Database Systems, Adv. Algorithm Concepts, Operating System Security, Theory of Computation

Bachelor of Technology, Information Technology

Charotar University of Science and Technology

Oct 2021 – May 2025

GPA: 3.29

Courses: Data Analytics & Visualization, Machine Learning, Python Programming & Hardware Interfacing, Database Management System, Data Structures & Algorithms, Advanced Computing, Software Engineering, Computer Networks

RESEARCH EXPERIENCE

IET Publishing — Forthcoming Book Chapter

July 2026

“A Calibrated Multimodal AI System for Real-Time, Treatment-Specific Mortality Prediction in the ICU”

- Designed a multimodal AI model using ResNet-50 and ClinicalBERT on **612K+ radiographs** from **42K+ ICU patients**, achieving **AUROC 0.91**.
- A prospective stepped-wedge cluster-randomized evaluation across 2,000 ICU patients showed potential for improving early risk detection and supporting timely treatment modifications through treatment-specific mortality prediction.
- Outputs treatment-specific counterfactual predictions, enabling clinicians to reason about how risk evolves under alternative fluid, oxygen, and vasopressor strategies in real time.

EXPERIENCE

Software Development Intern

SRKay Consulting Group

Dec 2024 – Apr 2025

- Lowered average design approval time by **35%** (sketch-to-CAD cycle benchmark) via exploratory data analysis and cleaning of **10,000+ records** in a multi-schema NPD database to surface workflow bottlenecks.
- Secured **100%** data quality pass rate across **68 referential-integrity steps** by building systematized scripts that catch inconsistencies at the source.
- Partnered with business stakeholders to translate technical findings into actionable recommendations, communicating analytical insights through SQL reports and Power BI dashboards.

Machine Learning Intern

Techno Smart Diamond Solutions

May 2024 – Jun 2024

- Engineered a CNN model using TensorFlow/Keras, attaining **95%** training accuracy and **100%** validation accuracy across 800 diverse images for a critical classification task.
- Scaled dataset **4x (200 → 800 images)** by developing an OpenCV pipeline for automatic image splitting and augmentation, eliminating manual labeling.
- Decreased validation loss from **0.66 to 0.03** (binary cross-entropy) using ImageDataGenerator augmentation, 80/20 split, and rigorous data cleaning.

PROJECTS

Diamond Part Recognition Model — Python, TensorFlow, Keras, OpenCV, Matplotlib

- Built a binary CNN classifier (Conv2D + MaxPooling, sigmoid output) reaching strong validation performance on diamond facet identification by training on 800 augmented images over 10 epochs with Adam and binary cross-entropy loss.
- Processed and analyzed unstructured image data using OpenCV, NumPy, and TensorFlow, expanding the dataset from 200 to 800 samples for model training.
- Deployed real-time prediction with an inference script that loads the .h5 model, normalizes inputs to 224×224, and outputs predicted labels with confidence scores for industrial robotics use.

New Product Development System — SQL Server, Python, Selenium, Cucumber, Excel

- Delivered descriptive analytics and performance reporting for **20+ designers**, using SQL pipelines across 15+ tables, surfacing data-driven insights that drove management adoption of a point-based scorecard.
- Reduced rework cycles by **20%** through root-cause analysis of rejected designs, identifying key bottleneck stages and presenting insights via Excel dashboards to business stakeholders.
- Orchestrated a streamlined approval system integrating design review checkpoints, cutting down design cycle time by **15%** and slashing rework requests from production by **12%** through early validation.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, C/C++, JavaScript, PHP

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, Keras, Deep Learning, Computer Vision, MLflow, Predictive Modeling, Statistical Analysis

Data Science & Big Data: Data Analytics, Data Engineering, ETL Pipelines, Hadoop, Hive, Apache Pig, MapReduce, HDFS, Distributed Computing

Analytics: Exploratory Data Analysis, Predictive Modeling, Insight Generation, Requirements Gathering

Libraries & Frameworks: NumPy, Pandas, OpenCV, Django, React.js, Flutter

Database: MySQL, SQL Server, MongoDB

Visualization & Analytics: Tableau, Power BI, Matplotlib, Seaborn, Excel

Cloud, Tools and Platforms: AWS, Docker, Git, Linux, Jenkins, Jupyter Notebook, VS Code, Wireshark, Figma